



SAPIENZA
UNIVERSITÀ DI ROMA

FOUNDATIONS OF DATA SCIENCE

Comprehensive Study Notes

Indro Spinelli

Department of Computer Science
Sapienza Università di Roma

NOTES PREPARED BY

Matt Clifford Trinidad Ramos

Calculus and linear algebra · Probability and statistics

Contents

I	Data Science and Machine Learning Foundations	1
1	What Is Data Science?	1
2	Machine Learning as a Subfield of AI	2
2.1	Tom Mitchell’s learning-task formulation	2
2.2	Why machine learning?	2
2.3	Machine learning and statistics	3
3	Learning Paradigms	3
4	Supervised Learning	3
5	Representative Supervised-Learning Tasks	3
5.1	Regression	4
5.2	Classification	4
5.3	A supervised model as a family of equations	6
5.4	Structured and high-dimensional outputs	6
5.5	Sequence and cross-modal generation	7
6	Unsupervised Learning and Generative Models	8
6.1	Latent variables	9
7	Reinforcement Learning	10
7.1	Chess as an example	10
7.2	Why reinforcement learning is difficult	10
8	Deep-Learning Landmarks	11
II	Data Representation and Linear Algebra Foundations	12
9	The Iris Dataset and Supervised-Learning Data	12
9.1	Features and target	12
9.2	Data types and numerical encoding	12
9.3	Labels and dataset notation	13
10	Data Generation and the i.i.d. Assumption	13
11	Data Points as Vectors	13
12	Spanning Sets and Bases	14
12.1	Spanning sets	14
12.2	Basis	14
13	Standard and Orthonormal Bases	15
13.1	The standard basis	15
13.2	Encoding with an orthonormal basis	16
14	Non-Standard Bases and Coordinate Transformations	16
15	Finding the Encoding Weights	17
16	Subspaces and Projection	18
16.1	Encoding, decoding, and projection	19
17	Quick Review and Terminology	21
	References	21

PART I

Data Science and Machine Learning Foundations

1 What Is Data Science?

Definition

Data science is an interdisciplinary field concerned with extracting actionable insight and knowledge from structured and unstructured data, using methods from statistics, mathematics, computer science, and domain expertise.

Prerequisites. Calculus and linear algebra (including derivatives and matrix–vector products), and basic probability and statistics (including Gaussian distributions, mean, and standard deviation).

Data science is broader than model fitting. A typical lifecycle starts from the problem that needs to be solved, passes through data and model development, and continues after deployment because models must be monitored and may need retraining.

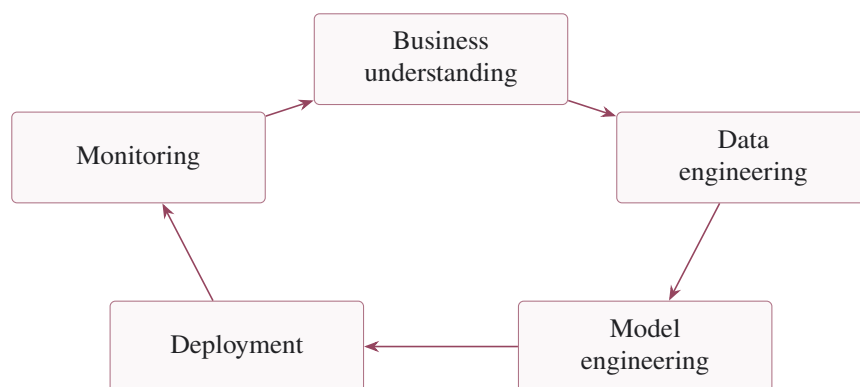


Figure 1. Data science lifecycle from business understanding through data engineering, model engineering, deployment, and monitoring. Feedback from monitoring initiates a new development cycle.

The stages involve the following activities:

- **Business understanding:** define business objectives and translate them into machine-learning objectives.
- **Data engineering:** collect and verify data, clean and visualize it, augment it when appropriate, and standardize it.
- **Model engineering:** select a model, train it, and evaluate it.
- **Deployment:** move the model into production and consider user acceptance and usability.
- **Monitoring:** monitor efficacy, collect new data, and retrain when required.

2 Machine Learning as a Subfield of AI

Artificial Intelligence (AI) is the field of computer science devoted to creating agents capable of tasks that typically require human intelligence, such as learning, reasoning, and problem solving. Supervised learning, unsupervised learning, and reinforcement learning are major machine-learning paradigms within AI. Deep learning cuts across these paradigms.

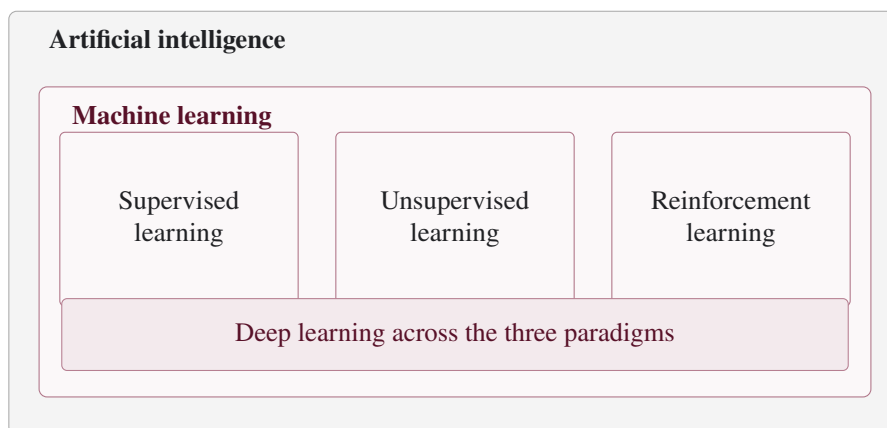


Figure 2. Artificial intelligence, machine learning, and the three learning paradigms. Deep learning supplies model families across the paradigms; the diagram is conceptual rather than a strict set-theoretic taxonomy.

2.1 Tom Mitchell’s learning-task formulation

In Tom Mitchell’s learning-task formulation, a machine-learning system improves its **performance** P at a **task** T through **experience** E . A well-defined learning problem can therefore be summarized by the triple

$$\langle P, T, E \rangle.$$

Notation. T denotes the task to be performed; P is the quantitative performance measure; E is the experience or data from which the system learns.

Table 1. Examples of tasks, performance measures, and learning experience.

Task T	Performance P	Experience E
Playing checkers	Percentage of games won against an arbitrary opponent	Practice games played against itself
Recognizing handwritten words	Percentage of words correctly classified	Human-labeled images of handwritten words
Driving on four-lane highways using vision sensors	Average distance traveled before a human-judged error	Images and steering commands recorded while observing a human driver
Classifying email as spam or legitimate	Percentage of messages correctly classified	A database of emails with human-provided labels

2.2 Why machine learning?

Many useful tasks are too complex to specify with explicit hand-written rules. Object recognition, speech understanding, and natural-language processing are canonical examples. Instead of manually encoding all rules, machine-learning systems learn useful regularities automatically from data or experience.

2.3 Machine learning and statistics

Statistics and machine learning share substantial mathematical and methodological foundations: both fields uncover patterns in data, rely on calculus, probability, and linear algebra, and share many algorithms. Their differences lie mainly in emphasis: statistics traditionally focuses more strongly on inference, conclusions, interpretability, and rigor, whereas machine learning often emphasizes prediction, scalability, and autonomy. These differences in goals and emphasis do not establish a hard boundary between the disciplines.

3 Learning Paradigms

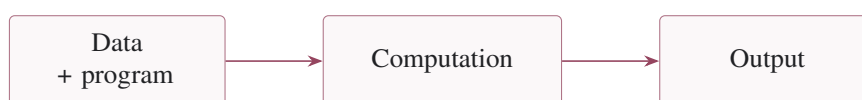
Table 2. Input/output view of the three paradigms.

Paradigm	Input	Output
Supervised learning	Examples of inputs paired with desired outputs	A model that predicts an output for a new input
Unsupervised learning	Examples of data without target outputs	A representation of structure in the data
Reinforcement learning	A sequence of interactions with an environment	A policy that performs a desired task

4 Supervised Learning

Supervised learning defines a mapping from input to output and learns this mapping from paired input/output examples. In traditional programming, *data + program* produce the output; in machine learning, *data + observed outputs* are used to learn the program/model.

Traditional programming



Machine learning

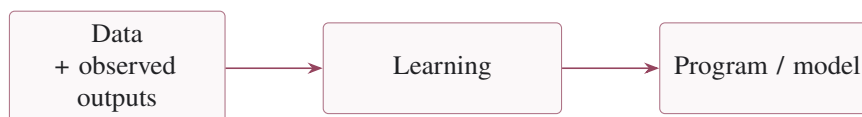


Figure 3. Traditional programming and machine learning: a program computes outputs, whereas learning uses observed inputs and outputs to construct a model.

5 Representative Supervised-Learning Tasks

A common prediction pipeline encodes raw real-world objects numerically, processes them with a model, and decodes the results into meaningful outputs. The examples differ primarily in the structure and dimensionality of the target.

5.1 Regression

Regression predicts real-valued quantities. In the univariate example, real-estate information is encoded as a feature vector and mapped to a single predicted price. In the graph-regression example, a molecular graph is encoded and mapped to more than one continuous target, namely physical properties such as freezing and boiling point.

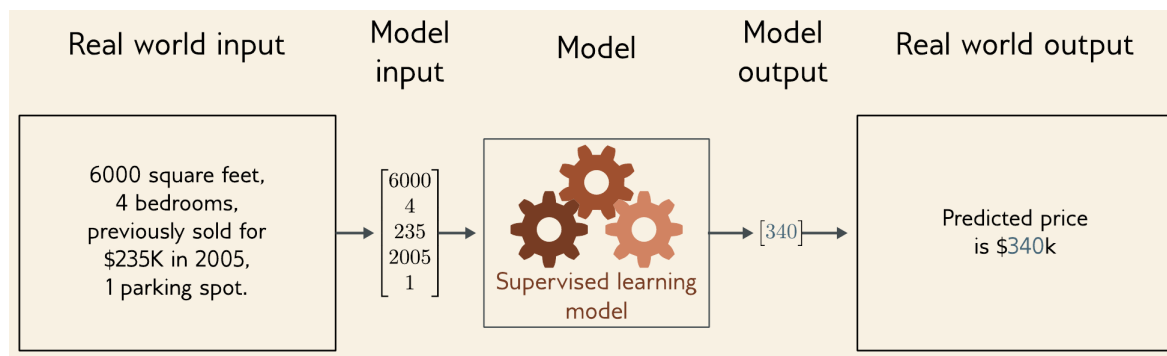


Figure 4. Univariate regression example: a real-world description is encoded numerically and mapped to a single real-valued output.

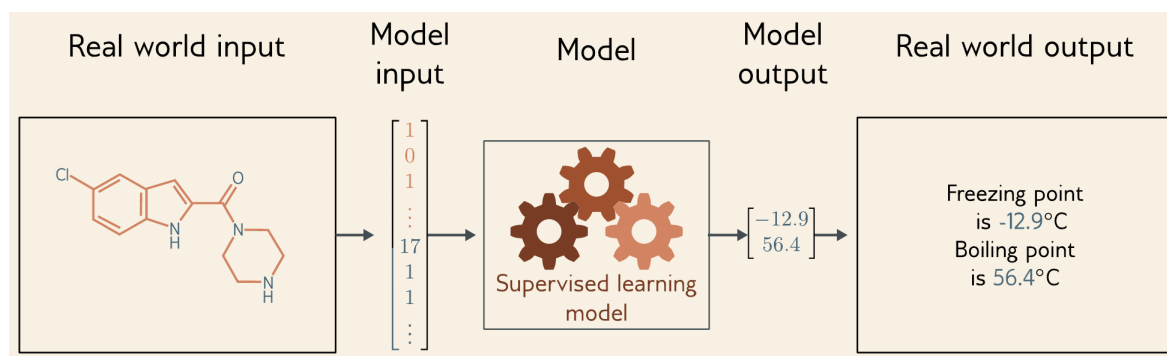


Figure 5. Graph regression example: a molecular structure is represented numerically and mapped to multiple continuous outputs.

5.2 Classification

Classification predicts discrete classes. A binary text-classification example assigns a sentence to one of two classes (positive/negative) using a Transformer. A music-genre example and an image-classification example illustrate multiclass prediction using, respectively, a recurrent neural network (RNN) and a convolutional network.

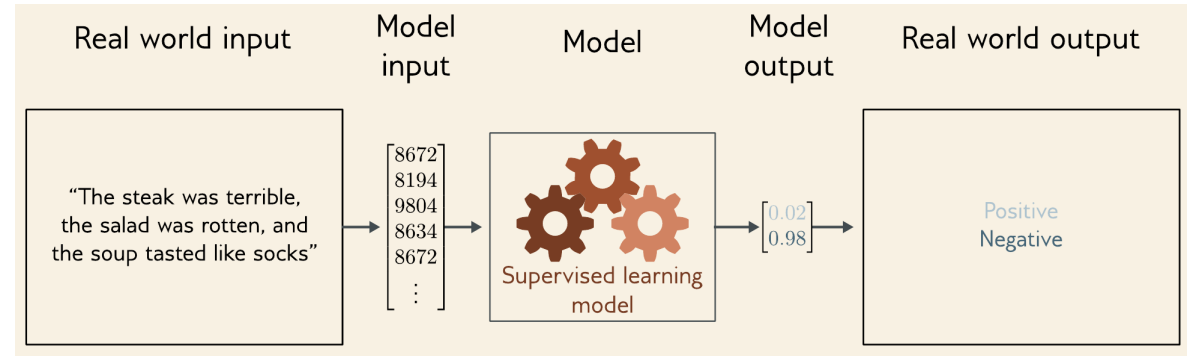


Figure 6. Binary text classification: text → numerical representation → class scores → positive/negative label.

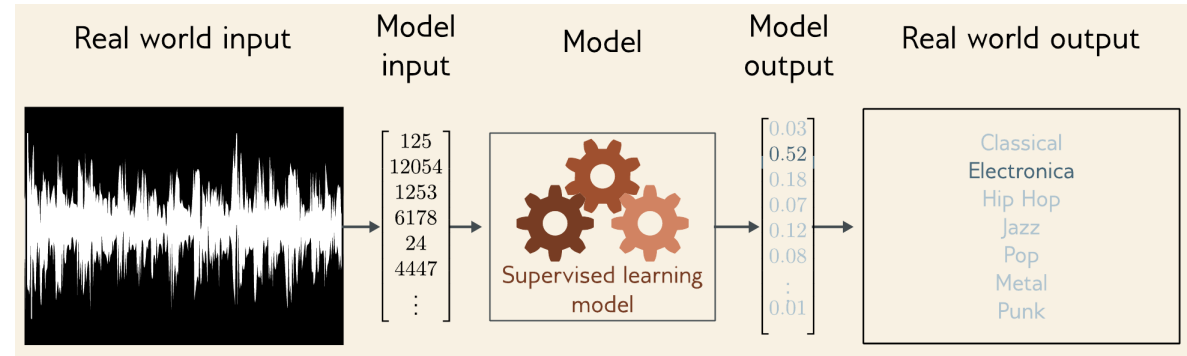


Figure 7. Multiclass music-genre classification using a recurrent neural network.

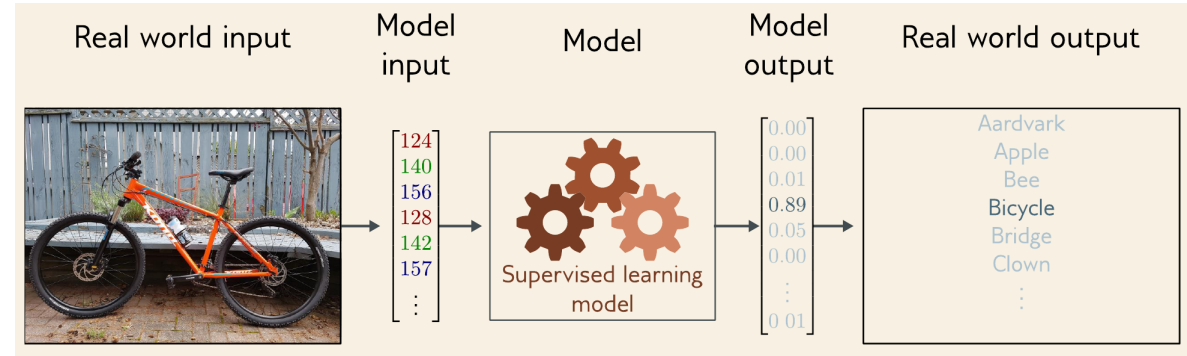


Figure 8. Multiclass image classification using a convolutional network.

5.3 A supervised model as a family of equations

A supervised model can predict height from age. Training means searching through a family of candidate equations and selecting one that fits the training observations well. Deep neural networks form extremely flexible families of equations; fitting such networks is an instance of **deep learning**.

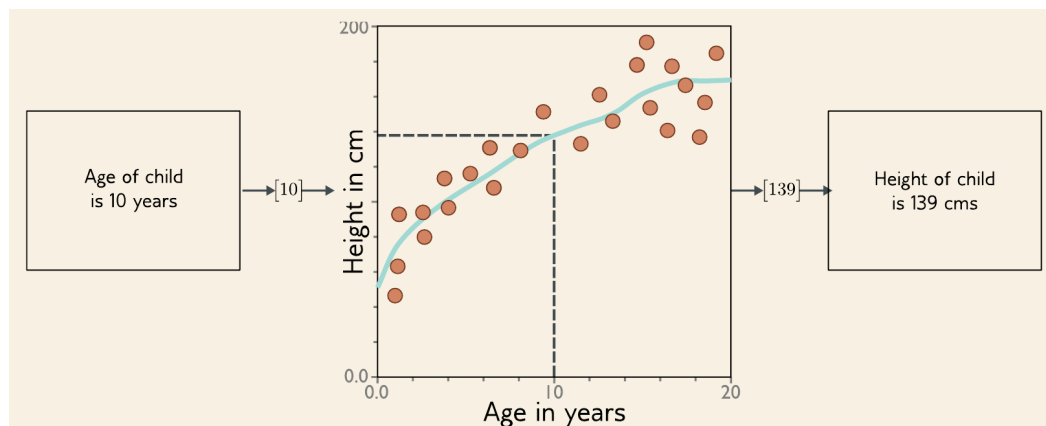


Figure 9. Supervised-learning view of model fitting: an input value is mapped to an output through an equation fitted to training data.

5.4 Structured and high-dimensional outputs

Some supervised tasks produce many outputs rather than a single scalar. Examples include:

- **Image segmentation:** many binary/discrete outputs, one per spatial location; the example uses a convolutional encoder–decoder network.
- **Depth estimation:** many continuous outputs forming a dense depth map; using a convolutional encoder–decoder.
- **Pose estimation:** many continuous outputs describing body/keypoint geometry; using a convolutional encoder–decoder.

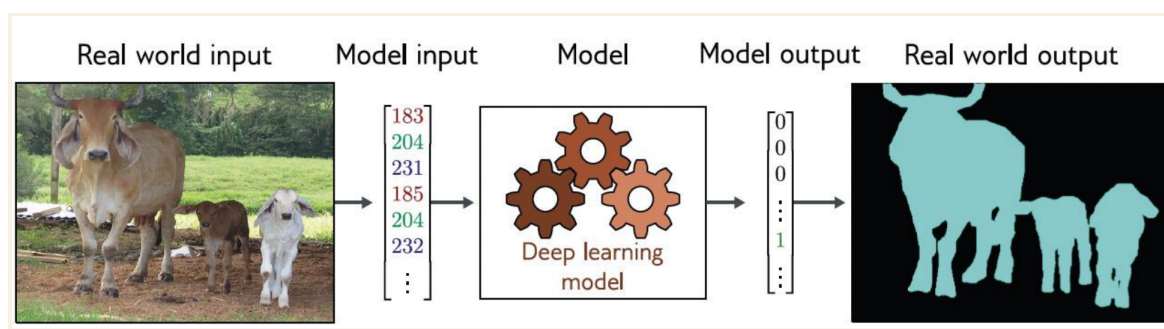


Figure 10. Image segmentation example: an input image is mapped to a dense discrete mask.

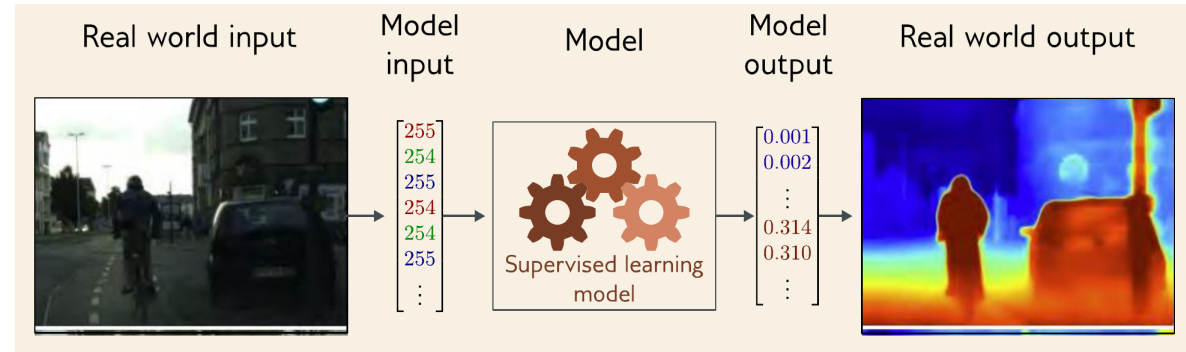


Figure 11. Depth estimation example: an input image is mapped to a dense continuous depth representation.

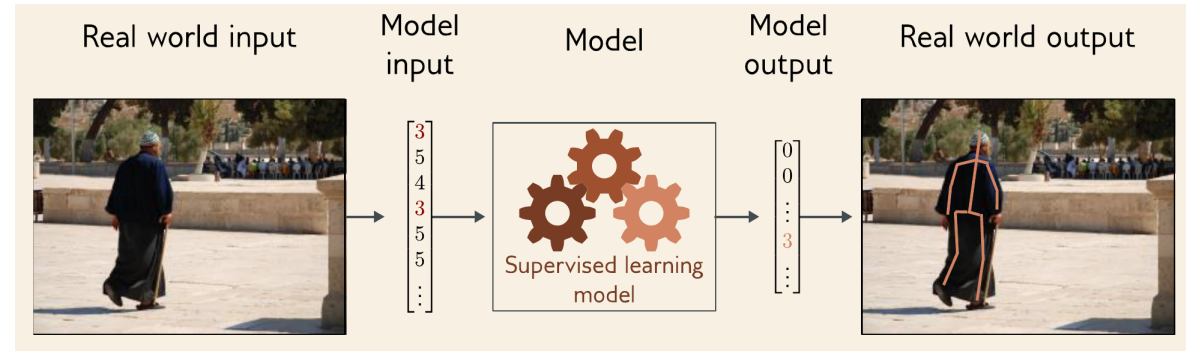


Figure 12. Pose estimation example: an input image is mapped to multiple continuous outputs representing pose.

5.5 Sequence and cross-modal generation

The same input–model–output pattern extends to structured sequences and multimodal tasks. Examples include machine translation (text sequence to text sequence), image captioning (image to text sequence), and text-to-image generation (text sequence to image).

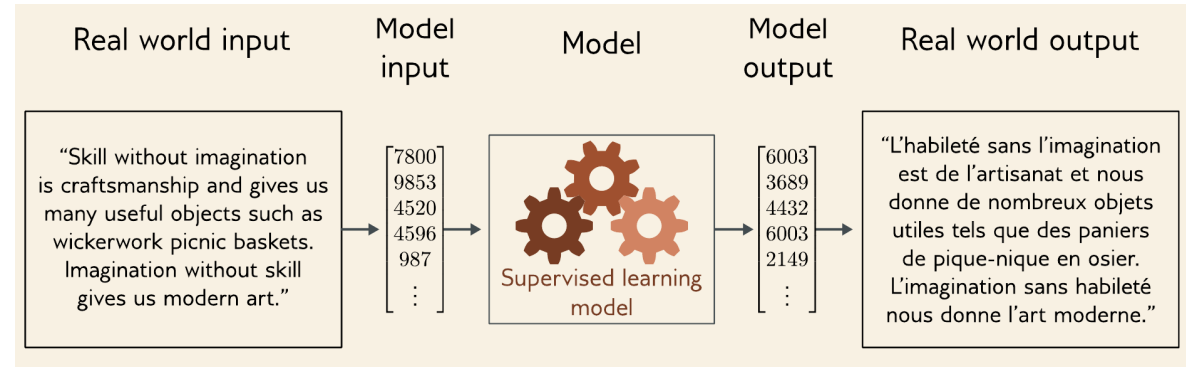


Figure 13. Machine-translation example: text is tokenized to a numerical representation and decoded into a target-language sequence.

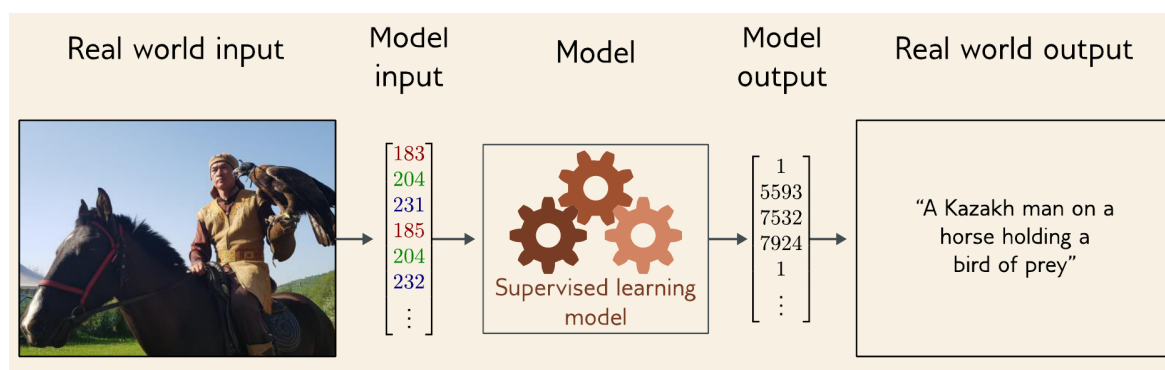


Figure 14. Image-captioning example: an image is encoded numerically and decoded into a textual description.

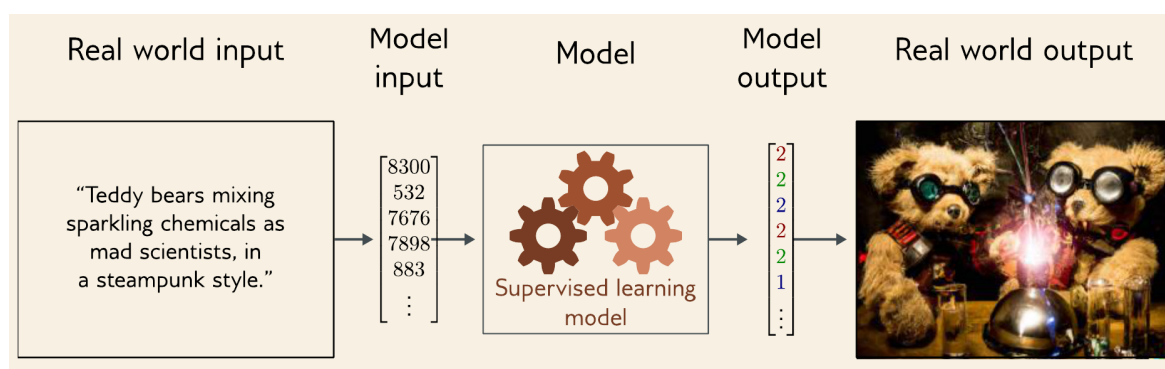


Figure 15. Text-to-image example: a text prompt is encoded and transformed into an image.

Key Concept

Across these examples, the relationship between input and output can be highly complex and there may be multiple valid outputs. Nevertheless, outputs—and often inputs—possess structure: language follows grammatical rules and natural images obey strong statistical regularities. This observation motivates learning the “grammar” of data from large quantities of unlabeled examples.

6 Unsupervised Learning and Generative Models

Unsupervised learning studies datasets without target labels. Representative goals include clustering, outlier detection, generation of new examples, and filling in missing data.

A central idea is to exploit the abundance of unlabeled data to learn structure before or alongside a supervised task. An example is *DeepCluster: Deep Clustering for Unsupervised Learning of Visual Features* (Caron et al., 2018), where visual examples are organized into coherent feature clusters without conventional class labels.

Generative models can create new examples. Approaches include generative adversarial networks (GANs), as well as probabilistic generative models such as variational autoencoders (VAEs), normalizing flows, and diffusion models.



Figure 16. DeepCluster: visually coherent groups discovered from unlabeled images. The panels illustrate visually coherent groups discovered from unlabeled images.

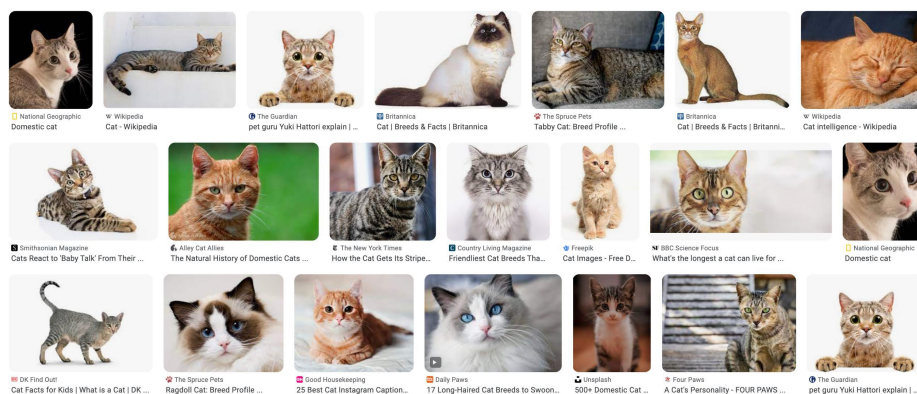


Figure 17. Examples of synthetic cat images produced by generative models.

6.1 Latent variables

A latent-variable generative model represents high-dimensional observed data through lower-dimensional latent variables. Samples are drawn in the latent space, transformed by the model, and decoded into observations.

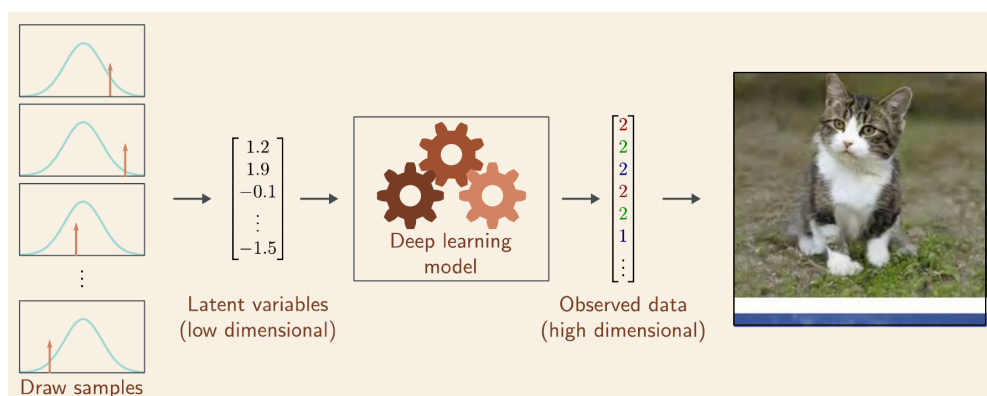


Figure 18. Latent-variable generation: low-dimensional latent samples are transformed by a deep model into high-dimensional observed data.

7 Reinforcement Learning

Reinforcement learning (RL) involves three basic ingredients: a set of **states**, a set of **actions**, and a set of **rewards**. The agent acts in the environment, changes state, and seeks behavior that produces reward. Unlike ordinary supervised learning, the agent is not given a fixed labeled dataset in advance; it must gather information through interaction.

7.1 Chess as an example

In chess, states are valid chess positions and available actions are valid moves. An illustrative reward scheme assigns positive rewards for taking pieces and negative rewards for losing them.

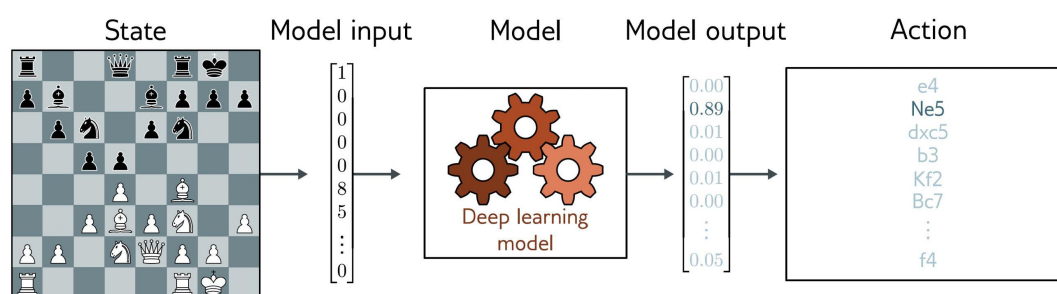


Figure 19. Reinforcement-learning pipeline for chess: board state $\rightarrow$ model representation $\rightarrow$ action scores $\rightarrow$ selected move.

Remark

Piece-capture rewards illustrate **reward shaping**: intermediate rewards provide feedback during the state–action–reward interaction loop.

7.2 Why reinforcement learning is difficult

Three central difficulties are:

- **Stochasticity:** the same action need not lead to the same future because other agents or the environment may behave differently.
- **Temporal credit assignment:** a reward observed now may be caused by an earlier sequence of decisions rather than the most recent action alone.
- **Exploration–exploitation trade-off:** an agent must decide whether to reuse a known good strategy or explore alternatives that might be better.

8 Deep-Learning Landmarks

Yoshua Bengio, Geoffrey Hinton, and Yann LeCun received the 2018 ACM A.M. Turing Award. The following timeline summarizes influential developments in neural networks and deep learning.

Table 3. Selected milestones in neural networks and deep learning.

Year	Milestone
1958	Perceptron (simple neural model)
1986	Backpropagation (practical deep neural networks)
1989	Convolutional networks (supervised learning)
2012	AlexNet image classification (supervised learning)
2014	Generative adversarial networks (unsupervised learning)
2014	Deep Q-learning for Atari games (reinforcement learning)
2016	AlphaGo (reinforcement learning)
2017	Machine translation (supervised learning)
2019	Language models (supervised/unsupervised learning)
2022	DALL-E 2, image synthesis from text prompts
2022	ChatGPT
2023	GPT-4 multimodal model
2024	Demis Hassabis and John Jumper: Nobel Prize in Chemistry in connection with AlphaFold-related work; Geoffrey Hinton and John Hopfield: Nobel Prize in Physics
2026	OpenAI announced an AI-generated proof concerning the Navier–Stokes Millennium problem; formal acceptance remains a separate matter.

Historical note (2 October 2026). OpenAI announced the proof on 8 September 2026; this announcement does not by itself establish a finalized Millennium Prize determination. See [OpenAI’s announcement](#) and the [Clay Mathematics Institute’s problem page](#).

Machine Learning Foundations — Key Takeaways

- Data science covers the full lifecycle from problem definition and data preparation to deployment and monitoring.
- Machine learning replaces explicit hand-coded rules with models learned from data or interaction.
- Supervised, unsupervised, and reinforcement learning differ primarily in the information available during learning and the object being learned.
- Deep learning supplies highly flexible model families that can be used across several learning paradigms.
- Modern tasks often involve structured, high-dimensional inputs and outputs rather than simple scalar prediction.

PART II

Data Representation and Linear Algebra Foundations

9 The Iris Dataset and Supervised-Learning Data

Fisher's Iris dataset provides a compact supervised-learning example. It contains **150 flowers**: 50 *setosa*, 50 *versicolor*, and 50 *virginica*. Each flower is described by four measurements in centimeters: sepal length, sepal width, petal length, and petal width. The target is the iris species.



Figure 20. The three Iris species: versicolor, virginica, and setosa. The classification task predicts the species from sepal and petal measurements.

9.1 Features and target

For n observed objects, the **features** are properties used to describe each object, while the **target** is the output variable or prediction goal. Supervised learning assumes that some relationship exists between features and target and attempts to learn a predictive rule from observed examples.

Table 4. Illustrative Iris data: four measurements in centimeters and the species label.

Sepal L.	Sepal W.	Petal L.	Petal W.	Species
4.3	3.0	1.1	0.1	setosa
5.0	3.3	1.4	0.2	setosa
7.7	3.8	6.7	2.2	virginica
5.5	2.5	4.0	1.3	versicolor

9.2 Data types and numerical encoding

Features and targets can have different types:

- numerical variables in $\mathbb{R}$;
- integer variables in $\mathbb{Z}$;
- categorical variables taking values in a finite set $\{C_1, \dots, C_g\}$;
- binary variables in $\{0, 1\}$.

For target variables, the distinction between continuous and discrete values leads naturally to regression and classification tasks.

Most learning algorithms expect numerical input. A categorical variable with k mutually exclusive categories can be represented by a one-hot vector

$$\mathbf{e}_i = (0, \dots, 0, \underbrace{1}_{i\text{-th position}}, 0, \dots, 0)^\top, \quad i \in \{1, \dots, k\}. \quad (1)$$

For categories with a natural order, an encoding should reflect that ordinality, for example through an ordered sequence of integers.

Notation. In equation (1), k is the number of categories and $\mathbf{e}_i$ is the binary indicator for category i ; exactly one coordinate equals 1.

9.3 Labels and dataset notation

Entries in the target column are called **labels**. Data are **labeled** when the target is observed and **unlabeled** when it is unknown.

Let X be the input space and Y the output/target space, with $X \subseteq \mathbb{R}^p$ in the typical vector-valued case. The dataset is written

$$D = \{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})\} \subset (X \times Y)^n. \quad (2)$$

Each observation satisfies

$$(\mathbf{x}^{(i)}, y^{(i)}) \in X \times Y, \quad i = 1, \dots, n, \quad (3)$$

and the j -th feature column can be represented as

$$\mathbf{x}_j = (x_j^{(1)}, \dots, x_j^{(n)})^\top \in \mathbb{R}^n. \quad (4)$$

Notation. n is the number of observations; p is the number of input features; $\mathbf{x}^{(i)}$ is the feature vector for observation i ; $y^{(i)}$ is its target; and $\mathbf{x}_j$ collects feature j across all n observations.

10 Data Generation and the i.i.d. Assumption

The observed dataset D is assumed to arise from an underlying probability distribution P_{XY} defined on $X \times Y$. The true distribution is generally unknown. Supervised machine learning seeks to learn enough of its structure to predict y from $\mathbf{x}$.

A standard assumption is that samples are drawn independently and identically distributed (i.i.d.) from P_{XY} . This contains two assumptions:

1. **Identically distributed:** every sample follows the same underlying distribution.
2. **Independent:** the realization of one observation does not depend on the other $n - 1$ observations.

This is a strong assumption and may fail in more complex settings such as time series.

11 Data Points as Vectors

For vector representations in the input space X , assume that the dataset has been **mean-centered**: the mean along each input dimension is subtracted so that the data cloud is centered around the origin. This operation is reversible as long as the feature means are retained.

Each observation can then be viewed as a point—equivalently, a vector—in a multidimensional vector space. A vector space is a collection of vectors closed under vector addition and scalar multiplication.

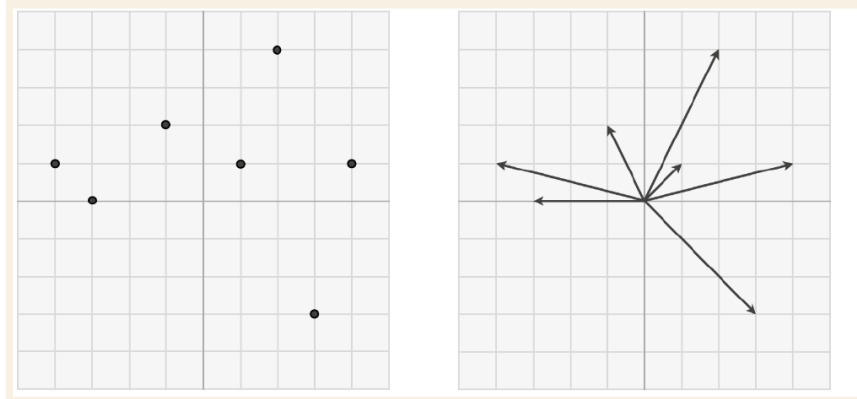


Figure 21. Two equivalent visualizations of the same 2-D data: points (left) and arrows from the origin (right).

12 Spanning Sets and Bases

12.1 Spanning sets

Definition

A **spanning set** is a collection of vectors whose linear combinations can produce every vector in the space being spanned.

In two dimensions, the standard vectors

$$\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

span the plane because any point $\begin{bmatrix} x & y \end{bmatrix}^\top$ can be written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = x\mathbf{e}_1 + y\mathbf{e}_2. \quad (5)$$

12.2 Basis

Definition

A **basis** is a linearly independent spanning set. In an N -dimensional vector space, a basis therefore contains exactly N vectors. A more general spanning set may contain more than N vectors, but then it is not a basis in the minimal sense.

For a candidate basis $\mathbf{c}_1, \dots, \mathbf{c}_N$ and each data point $\mathbf{x}_p$, perfect representation means that there are weights $w_{1,p}, \dots, w_{N,p}$ such that

$$\sum_{n=1}^N \mathbf{c}_n w_{n,p} = \mathbf{x}_p, \quad p = 1, \dots, P. \quad (6)$$

Notation. N is the ambient dimension; P is the number of data points; $\mathbf{c}_n$ is basis vector n ; $w_{n,p}$ is the coordinate/weight of data point p along $\mathbf{c}_n$; $\mathbf{x}_p$ is the original point.

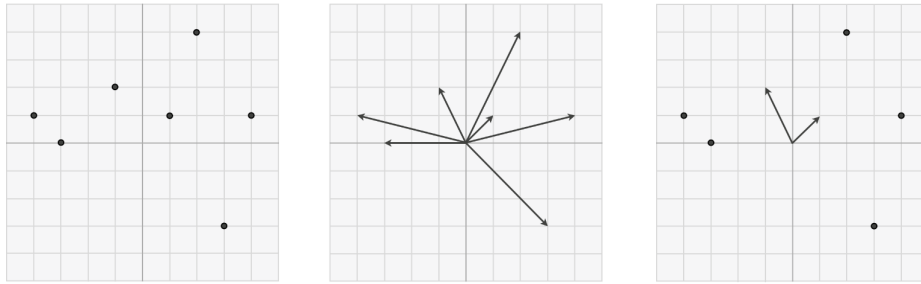


Figure 22. Points, vectors from the origin, and a basis representation in two dimensions.

13 Standard and Orthonormal Bases

13.1 The standard basis

The k -th standard basis vector has zeros everywhere except for a 1 in position k :

$$\mathbf{c}_k = (0, \dots, 0, 1, 0, \dots, 0)^\top.$$

For distinct basis vectors, the inner product is zero, while each vector has unit Euclidean norm:

$$\mathbf{c}_n^\top \mathbf{c}_m = 0, \quad n \neq m, \quad (7)$$

$$\|\mathbf{c}_n\|_2^2 = 1, \quad n = 1, \dots, N. \quad (8)$$

Thus the standard basis is orthonormal. If

$$C = [\mathbf{c}_1 \ \mathbf{c}_2 \ \cdots \ \mathbf{c}_N],$$

then

$$C^\top C = I_{N \times N}. \quad (9)$$

In the standard basis, the coordinate vector of a point is simply the point itself: $\mathbf{w}_p = \mathbf{x}_p$.

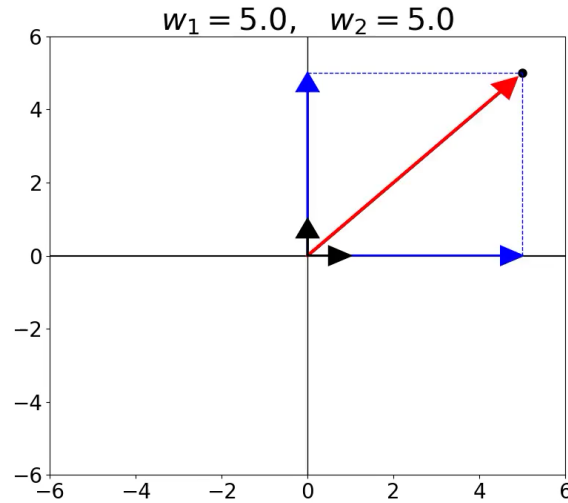


Figure 23. Reconstruction in the two-dimensional standard basis: a point $(5, 5)$ is reconstructed from weights $w_1 = 5$ and $w_2 = 5$.

13.2 Encoding with an orthonormal basis

For an orthonormal basis, the encoding of $\mathbf{x}_p$ is obtained directly by inner products:

$$\mathbf{w}_p = C^\top \mathbf{x}_p, \quad p = 1, \dots, P. \quad (10)$$

Reconstruction is exact when C is a complete $N \times N$ orthonormal basis:

$$CC^\top \mathbf{x}_p = \mathbf{x}_p, \quad p = 1, \dots, P. \quad (11)$$

Notation. C contains the orthonormal basis vectors as columns; $C^\top \mathbf{x}_p$ produces the coordinate vector $\mathbf{w}_p$; multiplying by C decodes the coordinates back into the original data space.

14 Non-Standard Bases and Coordinate Transformations

For a non-standard basis, the coordinate vector generally differs numerically from the original data point. The encoding is

$$\mathbf{w}_p = \begin{bmatrix} w_{1,p} \\ w_{2,p} \\ \vdots \\ w_{N,p} \end{bmatrix}, \quad (12)$$

which represents $\mathbf{x}_p$ in a transformed feature space whose coordinate axes are defined by the spanning vectors.

Consider the basis vectors

$$\mathbf{c}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}. \quad (13)$$

Because these vectors are linearly independent, any point in $\mathbb{R}^2$ can be expressed uniquely as

$$w_{1,p}\mathbf{c}_1 + w_{2,p}\mathbf{c}_2 = \mathbf{x}_p. \quad (14)$$

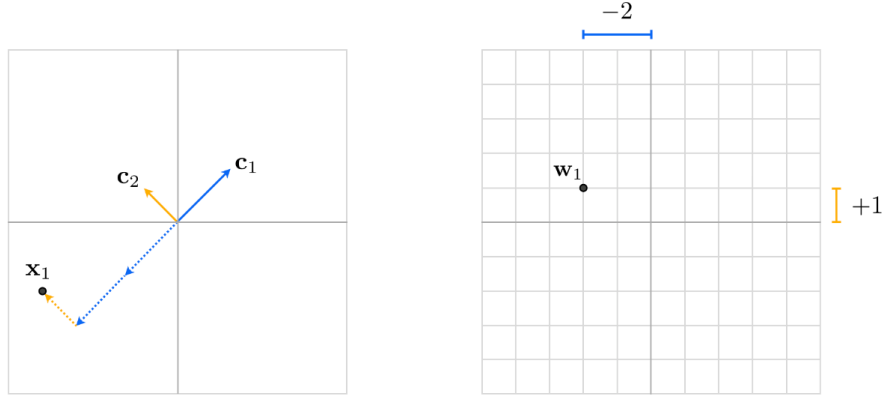


Figure 24. A point x_1 in the original coordinates (left) and its encoding w_1 in coordinates defined by the non-standard basis (right).

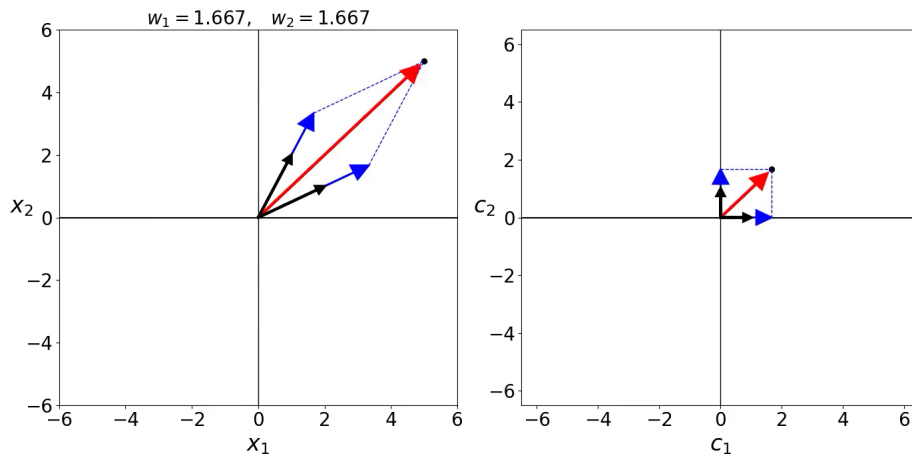


Figure 25. Geometric decomposition in a non-standard basis: the red vector is reconstructed from scaled basis vectors, while the coordinate vector records the required weights.

15 Finding the Encoding Weights

Let the basis/spanning vectors be stacked column-wise into

$$C = [c_1 \ c_2 \ \cdots \ c_N], \quad (15)$$

and let w_p contain the weights for point x_p . The squared reconstruction cost is

$$h(w_p) = \|Cw_p - x_p\|_2^2. \quad (16)$$

Setting the gradient to zero yields the symmetric first-order system

$$C^T C w_p = C^T x_p, \quad p = 1, \dots, P. \quad (17)$$

For all P points together, the optimization problem is

$$\underset{w_1, w_2, \dots, w_P}{\text{minimize}} \quad \frac{1}{P} \sum_{p=1}^P \|Cw_p - x_p\|_2^2. \quad (18)$$

Notation. C is the matrix of spanning vectors; $\mathbf{w}_p$ is the coordinate/encoding vector; $\mathbf{x}_p$ is the original point; $h(\mathbf{w}_p)$ is squared reconstruction error. Equation (17) is the normal-equation form obtained from the first-order optimality condition.

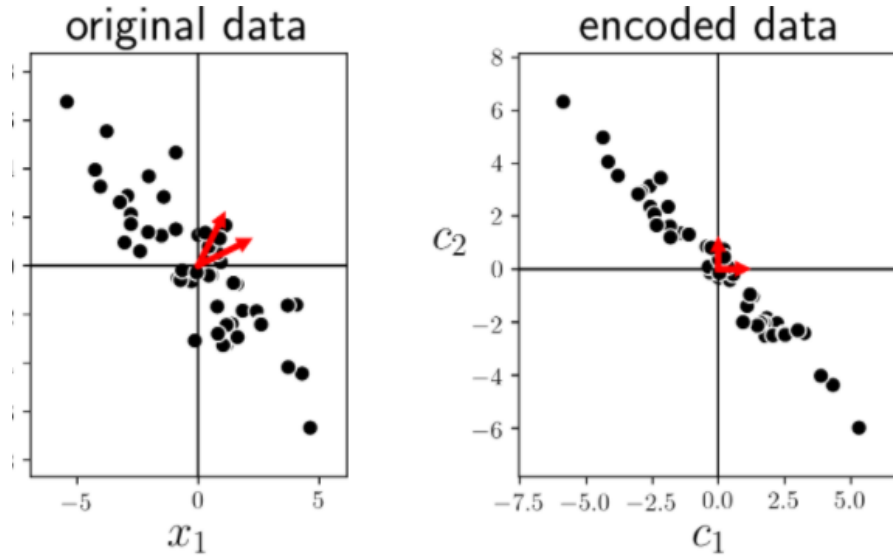


Figure 26. Original data versus the same observations represented in the coordinate system defined by a non-standard spanning set.

16 Subspaces and Projection

Up to this point, perfect representation in a generic N -dimensional space requires enough independent spanning directions. If only $K < N$ independent vectors are used, they can span at most a K -dimensional subspace of the full N -dimensional space. For example, in three dimensions, $K = 2$ independent vectors span a plane, while $K = 1$ spans a line.

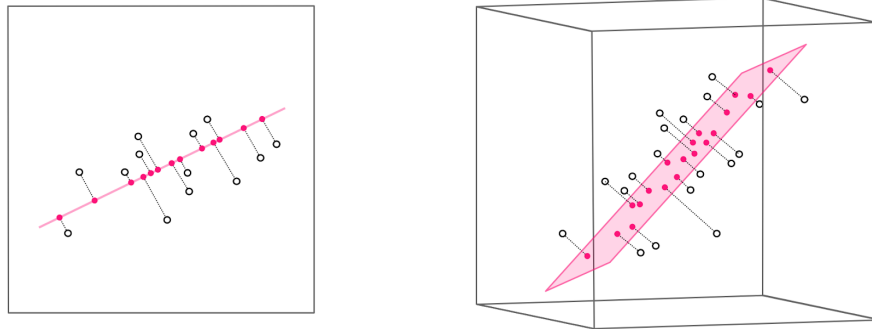


Figure 27. Lower-dimensional subspaces: a line or plane cannot represent every point in the ambient space exactly.

For K spanning vectors, the matrix now has shape

$$C = [\mathbf{c}_1 \cdots \mathbf{c}_K] \in \mathbb{R}^{N \times K},$$

but the first-order system keeps the same form:

$$C^\top C \mathbf{w}_p = C^\top \mathbf{x}_p, \quad p = 1, \dots, P. \quad (19)$$

16.1 Encoding, decoding, and projection

When $K < N$, the point cannot in general be reconstructed exactly. Instead, encoding maps the point to K coordinates, and decoding combines those coordinates with the K basis vectors to obtain the closest representable point in the subspace. Geometrically, this is an orthogonal **projection**: dropping the point perpendicularly onto the subspace.

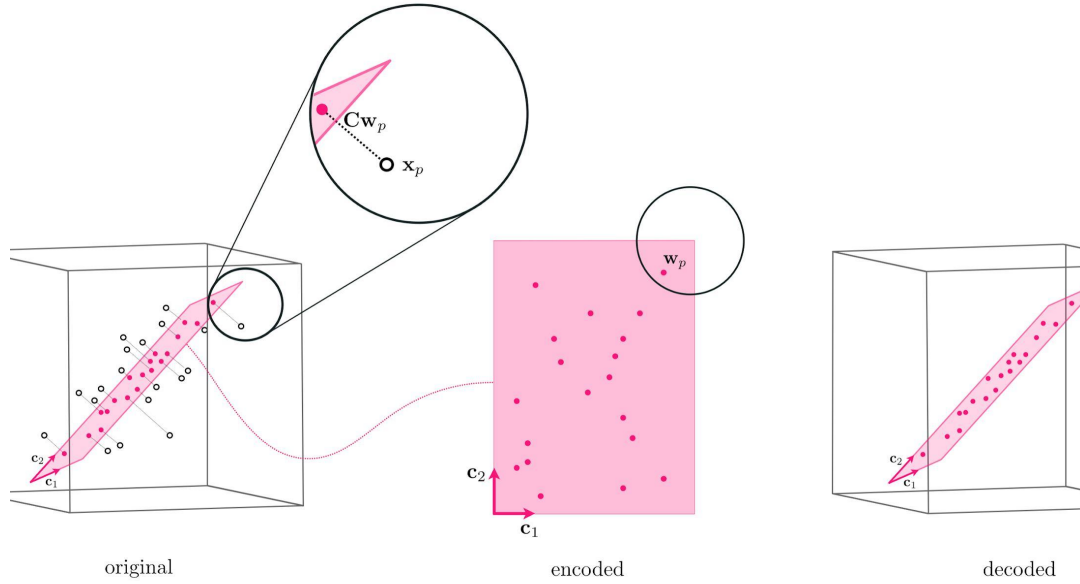


Figure 28. Encoding and decoding in a lower-dimensional subspace. The decoded point lies in the chosen lower-dimensional subspace and approximates the original point.

If the K subspace basis vectors are orthonormal, the encoding again simplifies to

$$\mathbf{w}_p = \mathbf{C}^\top \mathbf{x}_p, \quad p = 1, \dots, P, \quad (20)$$

and decoding gives the orthogonal projection

$$\mathbf{C}\mathbf{C}^\top \mathbf{x}_p \approx \mathbf{x}_p, \quad p = 1, \dots, P. \quad (21)$$

The approximation becomes equality only when the original point already lies in the spanned subspace (or when the orthonormal set is a complete basis of the whole space).

Notation. For an orthonormal $N \times K$ matrix $\mathbf{C}$, $\mathbf{C}^\top \mathbf{x}_p$ is the K -dimensional encoding and $\mathbf{C}\mathbf{C}^\top \mathbf{x}_p$ is the decoded projection in the original N -dimensional space. The residual $\|\mathbf{C}\mathbf{C}^\top \mathbf{x}_p - \mathbf{x}_p\|_2$ measures the projection/reconstruction error between the original point and its projection.

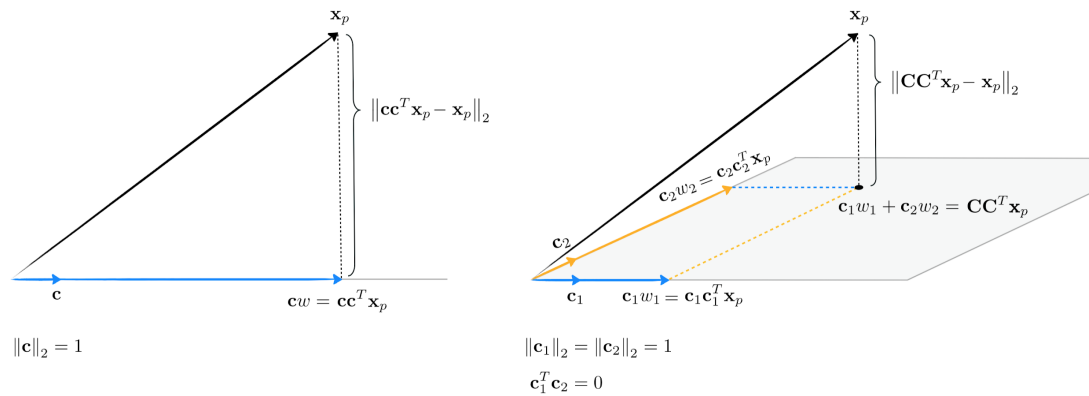


Figure 29. Geometric interpretation of orthonormal projection for one and two basis directions. The residual is perpendicular to the spanned subspace.

Vector Spaces and Projections — Key Takeaways

- A supervised dataset pairs feature vectors with targets; categorical features often require numerical encoding.
- Samples are modeled as independent, identically distributed draws from an unknown joint distribution P_{XY} .
- A spanning set generates a space through linear combinations; a basis is a minimal linearly independent spanning set.
- Orthonormal bases make encoding especially simple: coordinates are inner products, $\mathbf{w} = \mathbf{C}^T \mathbf{x}$.
- A non-standard basis changes coordinates but can still represent the same point exactly when it spans the entire space.
- With only $K < N$ directions, decoding produces a projection onto a lower-dimensional subspace rather than an exact reconstruction in general.

17 Quick Review and Terminology

Table 5. Core concepts in machine learning, data representation, and linear algebra.

Term	Meaning
Feature	Input property describing an observation
Target / label	Output variable to be predicted; a label is an observed target entry
One-hot encoding	k -dimensional binary representation with exactly one active coordinate
i.i.d.	Independent and identically distributed sampling assumption
Spanning set	Vectors whose linear combinations generate a space/subspace
Basis	Linearly independent spanning set; minimal representation of a vector space
Orthonormal basis	Basis vectors are mutually orthogonal and have unit norm
Encoding	Coordinate vector $\mathbf{w}$ representing a point relative to chosen spanning vectors
Decoding	Reconstruction $C\mathbf{w}$ in the original data space
Projection	Closest representable point in a lower-dimensional subspace, illustrated by $CC^T\mathbf{x}$ for an orthonormal subspace basis
Supervised learning	Learning a mapping from labeled input/output examples
Unsupervised learning	Learning structure from data without target labels
Reinforcement learning	Learning a policy through state/action/reward interaction

References

- [1] Deisenroth, Faisal, and Ong. *Mathematics for Machine Learning*.
- [2] Simon Prince. *Understanding Deep Learning*.
- [3] Watt, Borhani, and Katsaggelos. *Machine Learning Refined*, Chapter 8.
- [4] Caron et al. (2018). *DeepCluster: Deep Clustering for Unsupervised Learning of Visual Features*.
- [5] 3Blue1Brown. *Essence of Linear Algebra*, video series.